THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 4

Formal Systems and the Architecture of Proof

Exercises

Self-Study Curriculum

1. Consider the game of chess. Define it as a formal system: what is the language? What are the axioms (initial position)? What are the rules of inference (legal moves)? What constitutes a "theorem" (a reachable position)? Is the system complete, is every position either reachable or provably unreachable?

2. Euclid's fifth postulate (the parallel postulate) was controversial for two thousand years: it seemed less "self-evident" than the others. Eventually, it was shown that you can replace it with alternatives and get consistent geometries (hyperbolic, elliptic). What does this tell you about the relationship between axioms and truth?

3. In machine learning, a "model" is chosen (e.g., linear regression, neural network), then parameters are fit to data. In what sense is the choice of model analogous to the choice of axioms? What would it mean for a dataset to be "independent" of a model class, analogous to the Continuum Hypothesis being independent of ZFC?